/ Elie Galindo P.

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COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Chicago, Illinois 60607-7052

Quiz #4

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Open Notes and Book

Duration: 40 minutes

$A \rightarrow B$

$A \longrightarrow F$

Consider the irreversible conversion of species A into B and F through the reactions A → B and A → F in a stationary liquid film of thickness L. Determine C_A throughout the film, if the surface at y = L is assumed to be impermeable and unreactive, and the concentration of A at y = 0 is specified as C_o. The homogeneous reactions follow zeroth and first order kinetics with rates of reaction $R_{AB} = -k_B$ and $R_{AF} = -k_F \cdot C_A$, respectively. The diffusivity of the species A is D_A.

diffusivity $= D_A$

Find the overall rate of reactant consumption.

No flux at $y=L$

$A \rightarrow B \quad R_{AB} = -k_B$

$y=L$

$A \rightarrow F \quad R_{AF} = -k_F \cdot C_A$

Zeroth and 1st order kinetics

$y=0$ $\quad C_A = C_o$

$$R_A = R_{AB} + R_{AF} \rightarrow R_A = -k_B - k_F \cdot C_A$$

① - Defining boundary conditions (2 needed)

At $y=0 \rightarrow C_A = C_o$

②

At $y = L \rightarrow \left.\frac{dC_A}{dy}\right|_{y=L} = 0$

flux = 0 since impermeable and unreactive

At steady-state

non-homogeneos

Ficks law

$N_A = -D_A \frac{dC_A}{dy}$

Balance eq. $-\frac{d}{dy}(N_A)$

$0 = -\frac{dN_A}{dy} + R_A$

$\rightarrow 0 = \cancel{-}\frac{d}{dy}\left(\cancel{-}D_A \cdot \frac{dC_A}{dy}\right) + R_A \longrightarrow \underbrace{D_A \frac{d^2C_A}{dy} + R_A}$

$R_A = -K_b - \underline{K_F \cdot C_A}$

$D_A \frac{d^2C_A}{dy} - K_b - K_F C_A = 0$

-Dividing by D_A

$\frac{d^2C_A}{dy} - \frac{K_b}{D_A} - \frac{K_F \cdot C_A}{D_A} = 0$

Simplifying by using λ

$\lambda^2 = \frac{K_F}{d_A}$

$\frac{d^2C_A}{dy} - \frac{K_b}{D_A} - \lambda^2 C_A = 0$ or $\frac{d^2C_A}{dy} - \lambda^2 C_A - \frac{K_b}{D_A} = 0$

$C_A(y) = G\sinh(\lambda \cdot y) + F\cosh(\lambda y)$

$C_A(y) = G\sinh(\lambda y) + F\cosh(\lambda y) - \frac{K_b}{K_F}$

using the boundary conditions to solve for G and F

B.C ① At $y=0$ $C_A(0) = C_0$

$\cos(0) = 1$

$\sin(0) = 0$

$C_A(0) = G\sinh(0) + F\cosh(0) - \frac{K_b}{K_F}$

$C_0 = F - \frac{K_b}{K_F} \longrightarrow \underline{F = C_0 + \frac{K_b}{K_F}}$

B.C2 At $y=L$ $\left.\frac{dC_A}{dy}\right|_{y=L} = 0$

The derivative of $C_A(y) = G\sinh(\lambda y) + F\cosh(\lambda y) - \frac{K_b}{K_F}$

$$\frac{d(C_A(y))}{dy} = \lambda\left[G\cosh(\lambda y) + F\sinh(\lambda y)\right]$$

Using BC 2 — $y = L$

$$0 = \lambda\left[G\cosh(\lambda L) + F\sinh(\lambda L)\right]$$

$$G = -F\tanh(\lambda L)$$

$$C_A(y) = -\underline{F}\tanh(\lambda y)\cdot\sinh(\lambda y) + \underline{F}\cosh(\lambda y) - \frac{K_b}{K_F}$$

$$C_A(y) = F\left[\cosh(\lambda y) - \tanh(\lambda y)\sin(\lambda y)\right] - \frac{K_b}{K_F}$$

$$C_A(y) = \left(C_0 + \frac{K_b}{K_F}\right)\left(\cos(\lambda y) - \tan h(\lambda y)\sin(\lambda y)\right] - \frac{K_b}{K_F}$$

For the overall reactant consumption

$$N_A(0) = -D_A\left.\frac{dC_A}{dy}\right|_{y=0}$$

$$\left.\frac{dC_A}{dy}\right|_{y=0} = -\left(C_0 + \frac{K_b}{K_F}\right)\lambda\tanh(\lambda L)$$

$$N_A(0) = -D_A\left.\frac{dC_A}{dy}\right|_{y=0} \rightarrow D_A\left(C_0 + \frac{K_b}{K_F}\right)\lambda\tanh(\lambda L)$$

since $\lambda^2 = \frac{K_F}{D_A}$ or $\lambda = \sqrt{\frac{K_F}{D_A}}$

$$\text{overall rate of consumption} = \underline{\left(C_0 + \frac{K_b}{K_F}\right)\sqrt{D_A K_F}\tanh\left(L\sqrt{\frac{K_F}{D_A}}\right)}$$